



Latent tuberculosis: interaction of virulence factors in *Mycobacterium tuberculosis*

Sadhana Sundararajan¹ · Rajiniraja Muniyan¹

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Abstract

Tuberculosis (TB) remains a prominent health concern worldwide. Besides extensive research and vaccinations available, attempts to control the pandemic are cumbersome due to the complex physiology of *Mycobacterium tuberculosis* (*Mtb*). Alongside the emergence of drug-resistant TB, latent TB has worsened the condition. The tubercle bacilli are unusually behaved and successful with its strategies to modulate genes to evade host immune system and persist within macrophages. Under latent/unfavorable conditions, *Mtb* conceals itself from immune system and modulates its genes. Among many intracellular modulated genes, important are those involved in cell entry, fatty acid degradation, mycolic acid synthesis, phagosome acidification inhibition, inhibition of phagosome-lysosome complex and chaperon protein modulation. Though the study on these genes date back to early times of TB, an insight on their inter-relation within and to newly evolved genes are still required. This review focuses on the findings and discussions on these genes, possible mechanism, credibility as target for novel drugs and repurposed drugs and their interaction that enables *Mtb* in survival, pathogenesis, resistance and latency.

Keywords *Mycobacterium* · Target based medicine · Genetic interaction · Gene modulation · Antibiotic therapy · Dormancy

Introduction

Tuberculosis (TB), over the years have become an emerging infectious disease grabbing public health concern worldwide [1]. According to World Health Organization (WHO), TB is regarded to be leading in mortality rate. From 9 million incidences attributing to 0.9 million deaths in 2012, number of cases have gone up to 10 million with mortality rate of 1.2 million in 2019 [2]. After discovery of various drugs to treat TB, multi drug regimen approach was adapted for a faster and complete recovery. However, even more startling was the emergence of multi-drug resistant tuberculosis (MDR-TB) and extreme drug resistant tuberculosis (XDR-TB). MDR-TB is when the strain of *Mycobacterium tuberculosis* (*Mtb*) is resistant to rifampicin or isoniazid. XDR-TB is when the strain is multidrug resistant in addition to being

resistant to one fluoroquinolone class like ciprofloxacin and one injectable antibiotic like kanamycin, capreomycin and amikacin. Additional complications like drug-drug interactions also prevail for patients with complications like HIV, diabetes etc. [3, 4]. Early detection and complete utilization of prescribed antibiotics are important in containing the spread.

The severity of the disease and robust behavior of *Mtb* can be attributed its complex pathology, resilient cell wall and immune-evading strategies adapted to establish infection, persist and reactivate during favorable condition. The bacterium masks itself from the immune system by establishing modulations in its genes [5]. This review aims to emphasize the possible mechanisms of host-*Mtb* interaction, the genetic modulations responsible for cell wall synthesis, fatty acid degradation, iron uptake and oxidative stress resistance, their interactions and availability as targets for existing and proposed drugs are discussed.

✉ Rajiniraja Muniyan
rajiniraja.m@vit.ac.in
Sadhana Sundararajan
sadhana.s@vit.ac.in; sadhanasundararajan05@gmail.com

¹ School of Biosciences and Technology, Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India

Survival strategies of *Mtb*

Mycobacterium tuberculosis (*Mtb*) is an opportunistic slow-growing intracellular organism capable of latently affecting the host by corrupting the macrophages. Its multi-faceted virulence mechanism supports adhesion, accumulation, multiplication and infection at other sites. The alveolar macrophages (primary infection site) of lungs are activated to engulf the *Mtb* by bringing pro-inflammatory chemo-attractants like neutrophils, T-cells, interleukins (IL-1, IL-12) and NK cells promoting tissue remodeling [6, 7]. Normally, activated phagosomes fuse to lysosomes and its associated hydrolases. The phagolysosome complex digests the *Mycobacterial* contents using reactive oxygen and nitrogen species [8]. This event is restricted to the lungs where granulomas are formed along with immature macrophages, fibroblasts, lymphocytes and blood-derived monocytes surrounding the site of infection while eliminating the *Mtb* cells and prevent further tissue damage. However, *Mtb* have specialized mechanisms to evade the host strategies on apoptosis, like inhibiting the phagolysosome formation, phagosome acidification inhibition, expression of glycine-rich host protein and formation of an electron-transparent zone (ETZ) that impedes the diffusion of lysosomal enzymes [9].

Owing to low pH and oxygen in granuloma, it is proposed that *Mtb* is unable to survive. However dormant organisms may prevail for a considerable time. When the host immune system depletes due to conditions like HIV, immune suppressive drugs and aging, latent organism gain activity by depleting cell mediated immunity components like INF- γ or their secretors like T-cells as concluded in a mouse model [10]. One important strategy followed by macrophages to curtail *Mtb* is the secretion of inducible nitric oxide synthase (iNOS). *Mycobacterium* dampens the iNOS system by activating oxidative stress response genes like *ahpC*, *katG* [11]. Besides the perspectives on the pathogenicity of *Mtb*, the persistence, mechanism and latency remain a hotspot. Especially those regarding gene regulation for survival in host are rudimentary. However, *Mtb* does not have any typical virulence factors like toxins present in other bacteria. Thus it suggests the advanced adaptability of its genome to environment. This review provides collective information about genes responsible for the pathogenicity, resistance and survival of *Mtb* and deduces their interrelation with special emphasis on genes that draw attention as novel targets and factors that can be considered to resensitize *Mtb* to existing drugs.

Virulence factors for survival

It is interesting that *Mtb* lacks typical virulence factors that are common in other pathogens. In *Mtb*, virulence factors are classified as traits that are important for the progression

of TB. These factors include secretory proteins, cell wall component proteins, enzymes involved in lipid/fatty acid catabolism, amino acid biosynthesis, iron uptake regulation, oxidative stress response and transcription regulators. The importance and inter-relation of all such genes and their proteins are discussed in the section below.

Invasive ability

A group of highly conserved family of proteins encoded by *mce* (mammalian cell entry) gene group improves the adhesion, prolonged existence and invasion of *Mtb* into macrophages. They are primary lipid transporters across the hydrophobic cell wall. Some Mce proteins (especially Mce3E and Mce2E) decreases host cytokinin level by altering ERK1/2 (extracellular signal-regulated kinase 1/2) pathway. It blocks the ERK1/2 components from communicating with its upstream kinases MEK, resulting in downregulation of various cytokines like TNF and IL-6 [12, 13]. A PCR study that explained the penetrative ability of *Mtb* in non-phagocytic cells such as human anemone cells and HeLa cells, proved that these cells might act model to understand the physiology of pathogen uptake by mammalian cell [14]. It is shown that an *Mtb* fragment could turn non-pathogenic *E.coli* to invade mammalian cells [15]. While the strategies of phagocytic cells are explained, non-phagocytic cells affected in bacterial invasion are unknown [16]. A review, suggested that non-phagocytic cells may act reservoirs for latent *Mtb*, based on its ability to force entry into such cells [17].

There are four *mce* operons in the *Mtb* genome (*mce1-4*) that contain ~8–13 homologous genes arranged in identical manner with functions widely unknown. So far, best predicted function of *mce* genes are lipid metabolism and redox reactions [18]. Each *mce* loci comprises two integral membrane proteins *yrbE* (*yrbEA-B*) and 6 *mce* genes (*mce1-4 A-F*). Gene *mce1* whose main product is a 58 aminoacid membrane signaling/transporter protein along with other proteins from the operon is hypothesized for its involvement in host cell invasion by making changes in plasma membrane and improves sustainability. Gene *mce1R* acts intracellularly to repress the *mce1* operon, but the evidence of direct interaction is not found [19, 20]. The fatty acyl-CoA synthetase (*fadD5*) gene is also found as a part of *mce1* operon and its function is predicted at the initial stages of fatty acid degradation [19, 21]. The *mce1* operon is necessary to evoke host pro-inflammatory response for the establishment of an unflagging infection and it is largely controlled by numerous negative regulators or a global regulator [20], even at nutrient depleted conditions and stationary phase [22].

The *mce4* gene product is an ABC type trans-membrane cholesterol import system that enables *Mtb* to derive carbon and energy from the ubiquitous component of macrophage

membranes and works along with *fad* genes of *Mtb*, responsible for fatty acid degradation. Lack of *mce4* proved inefficient in acquiring cholesterol from host, as proven in mouse model [23]. A schematic representation of lipid import by *mce* and fatty acid synthesis are given in Fig. 1. Evidences show *mce3* gene can suppress β 2integrin-mediated signaling pathway and promote *Mtb* penetration into macrophages [24]. The *mce4A* homologs are highly conserved for some species of *Mycobacterium*. Function of gene products Mce3R (TetR family transcriptional regulator) and Mce2R are to repress *mce3* and *mce2* genes respectively. Additionally Mce3R negatively regulates many genes involved in lipid metabolism of *Mtb* [20]. The presence of *mce* operons in non-pathogenic *Mycobacterium* species drew the conclusion that the organism has acquired the gene, even before it became a pathogen and thus it does not limit its function in relation to expression. The Mce protein in mouse model showed that *mce1* gene product was virulent when investigated intraperitoneal implying that virulence might depend on the route of infection [25, 26]. The details on Mce proteins secreted by other genes of *mce* operon are scanty. Keeping in mind its small size, permeability, role in early infection sustainability and fatty acid import and catabolism, *mce* genes are proposed to be excellent for manipulation and hence form an eminent target for emerging drugs.

Lipid metabolism

During infection, *Mtb* sustains through varying stress conditions that alter its lipid composition. General perceptions are that host lipids act as primary carbon source for its survival. The trans-membrane Mce proteins are the cholesterol transport system needed in *Mtb* not essentially for infection but for persistence and long term survival [23]. It suggests that the coordination and expression of complete set of really complex genes like *fad*, *inhA*, *kasA/B*, *ksh*, *ahpC*, *cmA* (*mma*), *pca* etc are needed for lipid/fatty acid catabolism and cell wall mycolic acid synthesis as shown in Fig. 1.

Lipid catabolism

Abundance of host lipid, mainly cholesterol is important for *Mtb* persistence. There are > 80 genes encoding proteins for fatty acid degradation [27]. Cholesterol is absorbed by the ATP-dependent Mce4 transporter protein, metabolized into acetyl-CoA for utilization by TCA and glyoxylate shunt. Few enzymes like KshA/B, HsaC/D, KstD, KstA/B, FadA5, FadD3 are encoded by 2 adjacent *fadE* genes involved in cleaving the aromatic cholesterol rings A and B, regulated by cholesterol itself [27, 28]. 3-ketosteroid 9 α hydroxylase

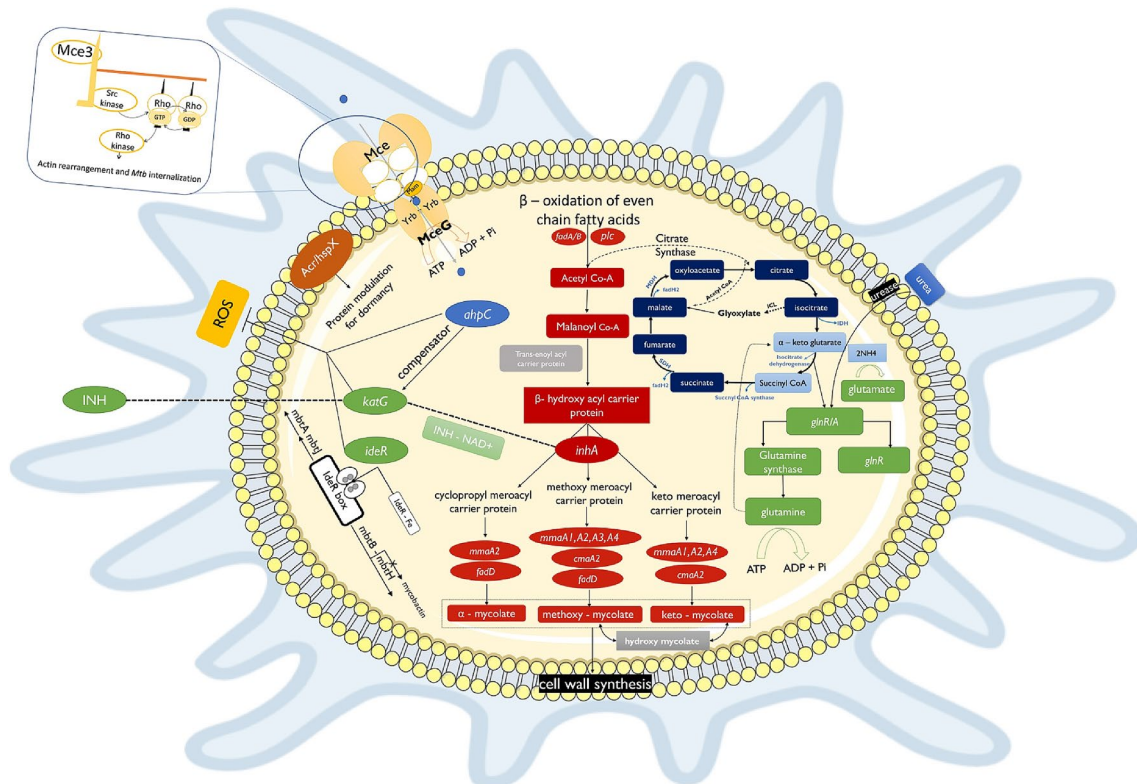


Fig. 1 Schematic representation of possible virulence gene interaction in *Mycobacterium tuberculosis*

(Ksh) is a 2 component oxygenase-reductase complex where KshA is an oxygenase and KshB is a reductase.

The ringed cholesterol structure of macrophages has 5-carbon side chain; steroid cyclohexane rings A, B, C and a cyclopentane D ring. Cholesterol catabolism begins from the side chains while it is connected to the steroid rings. This action is performed by *fadE26/27* gene. Because of its function, it is now renamed as *ChsE1/2* (cholesterol side chain metabolism) [27]. Cholesterol degradation yields propionyl-CoA that is converted to methyl-malanoyl-CoA, on combining with CO₂. Methyl-malanoyl-CoA is a metabolite in methyl branched lipid biosynthesis.

Ksh participates in the steroid B ring opening. Partial disruption of Ksh protein was comparatively found to impair the cholesterol side chain breaking due to accumulation of Co-A thioesterase in a mutagenesis study [29]. Similar studies were done in-vivo and in-vitro deleting *kshB* resulted in quick changes in cell wall with no marked changes in physiology [30]. Successively KstD exhaustively catabolises the steroid A ring to produce CO₂, providing energy that is assimilated by the TCA/Glyoxylate shunt. The B ring is broken down to its carbon components taken by lipid pool. *Mtb* takes up androstenedione/androstadienedione pathway (AD/ADD) using KstD, which initiates steroid ring degradation in a substrate specific manner [31]. The *kstR1/2* regulons control 4 of the 6 genes above mentioned in a TetR type regulation. However, *kstR2* regulation of the *fadE31-33* gene cluster are yet to be validated [29].

The *igr* operon contains multiple copies of genes that are involved in β -oxidation of cholesterol and are regulated by 4 genes (*sigE*, *kstR1* (also *kstR*), *kstR2* and *mce3R*). Early studies on *igr* operon knockout, resulted in lesser cholesterol uptake, confirming the importance of the genes under its control [29, 32]. Among the 4 regulatory genes, the role of *mce* in cholesterol import proved to be critical in many cases. In mice, *mce4* gene mutant showed disrupted cholesterol transport into the *Mtb* cell [23]. A set of *fadE* genes (*fadE17* and *fadE18*) that aims in degrading carbon side chains, are instigated by cholesterol and repressed by *mce3R* [23, 32]. In a study attempting to isolate and characterize FadE17-18, protein, concluded its structure as a 2- α ,2- β heterotetramer by bioinformatic analysis. However, isolation of soluble complex of FadE17-18 was unsuccessful, while those of FadE23-24 and FadE26-27 were isolated successfully. It was concluded that FadE26-27 had two active sites that participates in cholesterol side chain dehydrogenation. On attempting to complex the triplet FadE31-FadE32-FadE33, only FadE31 did not yield to be soluble [27]. In conclusion, fatty acid metabolism-related gene expressions reflect a common pattern responsible for fundamental adaptation to intracellular growth. It serves as the converging point to genes controlling fatty acid metabolism, cell wall integrity and reactive oxygen species detoxification. Hence,

fad gene down-regulation can be an effective way to combat dormant organisms.

Mycolic acid biosynthesis

Another gene responsible for *Mtb* survival is the *katG* gene encoding catalase peroxidase—a known target for isoniazid (INH) (iso-nicotinic acid) and ethambutol. Isoniazid was successful in practice from 1952 until a few years later when drug resistant TB arose. The *inhA* participates in cell wall mycolic acid synthesis along with *fad* and *cma* genes enabling *Mtb* to survive in adverse conditions attaining latency. FAS-II (Fatty Acid Synthase) system takes up the medium sized acyl-CoAs from FAS-I system as primers, condenses it with melonyl acyl carrier protein (ACP) [33]. The InhA is a short chain dehydrogenase that reduces, β -Keto acyl ACP product from the preceding step in an NAD(H) dependant manner. The successive condensation steps are done by β -Keto acyl ACP synthase (*kasA* and *kasB*) resulting in very long chain meromycoyl ACPs that are precursors for mycolic acid [34] as shown in Figure 1.

INH inhibits long chain fatty acid synthesis by targeting the enoyl acyl carrier protein (InhA) and KasA. The *inhA* is interrupted by the adduct formation between NAD(H) and INH which also involves the catalase peroxidase (KatG) by forming an iso-nicotonyl radicle [35]. So theoretically, *inhA* should be inhibited in sensitive organism and expressed in resistant strain of *Mtb*. Early studies have reported the sensitization of INH resistant *Mtb*, with the assistance of functional catalase [36]. Hence assumptions on *katG* involvement in INH activation/conversion to active intermediate prevailed.

Resistance may result from mutations in specific sites of the target gene corresponding to drug activating protein. Highly resistant *Mtb* strains lacked *katG* gene, compelling the evidence of catalase for drug uptake and virulence. Biochemical and immunopathological studies on clinical samples concluded the same [37, 38]. Among the aberrations in *katG* gene, one unique representation was found to be Ser315 or Arg46 substitution mutation. Initial conception of resistance was large deletions/entire gene deletions. However, later it was experimentally proven that simple deletions, mis-sense or terminal mutations were responsible for the loss of activity of *katG* [39, 40]. It is interesting that ~85% of rifampicin resistant strains are also resistant to INH. Rifampicin resistance confers to Ser531Leu, a common point mutation in specific 69bp RRDR (rifampicin resistance determining region) in the *rpoB* (encoding RNA polymerase β subunit) gene [41, 42]. Another common *katG* mutation observed was the Arg463Leu, which however did not considerably affect the conversion of INH to iso-nicotinic acid as shown in studies [43]. Thus it drew attention to investigate the drug elimination mechanism of genes co-existing in lipid

metabolism and cell wall modification of *Mtb*, such as *katG*, *inhA* and *ahpC*.

Some studies suggested the over expression of efflux pumps genes like *mmpL3*, *mmpL7*, *efpA*, *mmr* and genes like *ahpC*, *kasA*, *ndh* along with mutations/deletions in *katG* contribute to collective resistance towards first line antibiotics [44, 45]. *Mycobacterial* membrane proteins large (*mmpL*) translocates complex virulence associated envelope lipids and siderophores across the plasma membrane to periplasmic space. Most of the *mmpL* class proteins are predicted to be > 100kDa with a few exceptions [46, 47]. It indirectly enables *Mtb* to survive in host at initial exposure to antibiotics. However, complete functioning of genes like *mabA*, *mmpl*, *mmr* that are involved in the chain are yet to be completely studied [48]. It is hypothesized that when *katG* and INH synergize to create toxicity in *Mtb*, the *ahpC* and *ideR* might play a detoxifying role (Fig. 1). In case of mutational loss in *katG*, *ahpC* would play a substitution role [1]. Since *katG*, *rpoB*, *ahpC* and *mmpl* are interrelated in sustaining the organism at early stages of infection, targeting one of the genes could increase the scope of curtailing the infection. Hence these genes form attractive targets to candidate/proposed drugs.

The *Mtb* relies upon the hydrophobic cyclopropanated cell wall mycolic acids, to escape the oxidative stress imposed by the macrophages. Alkyl hydroperoxidase is encoded by *ahpC* to escape the deleterious effects of reactive oxygen and nitrogen intermediates and in cases compensate the mutational loss of *katG* to improve resistance to antibiotics. It reduces a variety of peroxides into its corresponding alcohols. The lipid rich cell wall has drawn interest as drug target for its extremely low permeability [3, 49].

The *ahpC* is tightly related to *oxyR* gene despite many inbuilt mutations, it positively regulates at least 9 genes including *ahpC* [50]. A mutation in *oxyR/ahpC* confers resistance to INH. It is evident from the study conducted by Ramos et al., (2018) that, expression of *ahpC* could be hindered completely by inactivating *oxyR* [51]. Nevertheless there was no loss of virulence *in-vivo* and in human macrophages. Some reports suggest that *ahpC* works tight with *oxyR* or *oxyR* is differently transcribed from *ahpC* and regulates *ahpC* itself, but complete single contagious *ahpC* product is not defined due to multiple lesion interruptions in sequenced *ahpC* gene, confirmed by whole genome sequencing [51, 52]. It is recently reviewed that *ahpC–oxyR* intergenic (specific 109 bp region) accounts the most number of mutations in addition to *katG* and *inhA* mutations contributing widely to INH resistance [53]. A DNA analysis involving ~ 1000 clinical isolates concluded that *ahpC* is most likely involved in rifampicin and INH resistance. They postulated that resistance developed to one drug might increase the probability of developing resistance to other drugs as well. However, this supposition was based on the

consideration that *ahpC* mutation as a single factor is not sufficient for INH resistance, yet found along with *katG* activity [54]. In a pyrosequencing study, specific *Mtb* gene regions (*katG*, *inhA*, *ahpC*, *rpoB*, *gyrA*, and *rrs*) in ~ 1128 clinical specimens, determined 10% *ahpC* mutations including its promoter for the detection of INH resistance [55]. Direct sequencing and PCR analysis of 9 genes supposedly involved in resistance to anti-TB drugs and latency, indicated at least 10% of the isolates had *ahpC–oxyR* mutations [56, 57]. A cross-resistance determination study deduced the relation between INH resistant strains with mutations in genes *katG*, *inhA* and *ahpC–oxyR* intergenic region had little influence to pasiniazid resistance suggesting pasiniazid may be an effective alternative to INH (para-aminosalicylic acid- derivative of INH) [58].

The *Mtb* has covalently linked arabinogalactan-mycolate complex attached to peptidoglycan of the cell wall through phosphodiester bonding. Trehalose dimycolate in the cell wall also supports structural stability. Mycolic acids (MA) are high-molecular weight β -hydroxy fatty acids with long α -alkyl side chain forming the key component of the cell wall. Alongside this, some free fatty acids are also complement long and short α -chains [33]. There are 3 structurally distinct classes of mycolic acid generated by *Mtb*, alpha-, keto- and methoxy mycolic acids. *M.smegmatis* behaves slightly different with respect to generation of a special mycolic acid unit called hydroxy mycolate or epoxy mycolate [59]. It is an early perception that epoxy mycolate are modulated α -mycolic acids through methyl transferase (coded by *cmaA22* or *mma4* gene) mediated double bond cleavage, end product being hydroxy mycolate. Based on proof of genetic similarity in critical aminoacids, *Mtb cmaA* gene was cloned in *M.smegmatis*. It produced both keto- and hydroxy- mycolates [60].

Interestingly, hydro-mycolate was considered a common intermediate for keto mycolate and methoxy mycolate and required *cmaA2* (*mmaA4*) for synthesis. Further down this lane, mutated *cmaA2* strain showed defects in trans-ring on the methoxy-mycolate and keto-mycolate. Other genes of the same family including *pcaA* (proximal cyclopropane synthase of a mycolates), *mmaA2*, *cmaA221*, *cmaA222*, *umaA1*, *mmaA1*, *mmaA4* and *mmaA3* prevail with striking genetic similarity [61, 62]. Gene *cmaB* (*mma3*) encodes the protein for methoxy property of mycolate, *cmaC* (*mma2*) and *cmaD* (*mma1*) belongs to the same gene cluster, involving in the cyclopropane forming function of mycolic acids. *In-vivo* testing on mouse model with inactivated *cmaA* and *mmaA4* revealed an attenuated phenotype, indicating that oxygenated mycolic acids are essential in infection process [62]. Additional symptoms such as aggregated immune response leading to excessive tissue damage (IL-12 production) were seen *in-vivo* [63]. In some cases, *pcaA*, a novel member of cyclopropane synthetases family was concluded essential for

pathogenesis when a *pcaA* mutant could not initiate persistent infection *in-vivo*. It is postulated that *pcaA* is needed in α -mycolate for cyclopropane ring formation. Inactivation of *pcaA* by gene silencing affected the cyclopropanation of α -mycolate that confirmed its importance. This finding was contradicting to their previous views that this function belongs to *cmaA2* [61, 64]. They also have noted an 8 amino acid segment specific to *cmaA2* and concluded that this segment might be important for its cyclopropanation activity. It is notable that all genes in the mycolic acid synthesis are cis-cyclopropane synthetase except *cmaA2* gene [61, 65]. Keto mycolic acids are required for organizing the organism in a pellicle that confers to drug resistance and latency that are otherwise sensitive. Regarding biofilm production, it is still unclear if the organism prepares true biofilms in human host or it is the action of latent tuberculae [66]. Although aspects of mycolic acid synthesis and transport are unveiled deep, many pathways are still not understood clearly. The problem with *cma* or *pca* as drug targets are that, point mutations in these genes can make them resistant to new drugs. Also, targeting *cma* gene as a whole entity, might be deleterious to the host. Nevertheless, proteins/genes for mycolic acid synthesis are excellent targets to improve cell permeability when inhibitors with broad activity against mycolic acid precursors are given as drugs rather than targeting a single gene/protein.

Energy source for latent *Mtb*

As a hetrotroph, *Mtb* is believed to switch between active and dormant stages based on the availability of multiple carbon sources by alternating metabolic pathways. The Icl (Iso citrate lyase) enzyme switches between TCA cycle and glyoxylate shunt by acting as isocitrate lyase and methyl-isocitrate lyase to breakdown propionyl-CoA (Product of odd-chain fattyacid degradation) [67]. Other genes like *plc* that hydrolyses phospholipids of host cell membrane [68], *gln* that performs ammonia assimilation and glutamine synthesis thereby inhibiting phagosome acidification [69] are associated genes that influence *Mtb* survival.

The *Mtb* has 4 gene clusters (*plcA-D*) that encodes Plc protein that hydrolyses phospholipids, release fatty acid that are used in β -oxidation cycle and glyoxylate shunt (Fig. 1). The whole genome sequence of *Mtb* revealed that *plcA-C* are full-length genes and *plcD* is truncated and located in a different region. The classification is based on their site of cleavage. Though PhospholipaseD (*pld*) and *plc* are two separate genes encoding phospholipase, *plc* (especially subunit C of *plc*- provides nutrients like fatty acids) is restricted to virulence activity in *Mtb* [68, 70].

Early detection studies in infected macrophages showed increases *plc* activity. RT-PCR transcripts of *plc* concluded strong gene upregulation during initial macrophage

infection and induces cell necrosis [71, 72]. Additionally, *plc* may degrade the phagosomal membrane by generating inositol-3-phosphate (IP3) and diacetyl-glycerol (DAG) that acts as secondary messengers. DAG is Ca^{2+} dependent and a physiological activator of protein kinase C (PKC) that in turn activates *pld* producing more of DAG. It thus builds a positive feedback mechanism which increases the intracellular Ca^{2+} content [73]. It is postulated that calcineurin dependant Ca^{2+} influx blocks phagosome-lysosome fusion by an unknown mechanism [9, 74]. Attempt to isolate and characterize *plc* gene are in vain because of the sparse quantity of Plc proteins. Recombinant-Plc proteins from *M. smegmatis* seemed to aggregate macrophage infection faster *in-vivo*. Recent studies on novel proteins with similar phospholipase activity have been identified [75, 76]. Thin-layer chromatography (TLC) analysis indicated presence of *plc* and additional *pld* activities. The genes *plcA* and *plcB* (also designated *mpcA* and *mpcB*), are arrayed in the same gene separated by 191 bp [77]. Integrated comparative genomic study, deduced the relationship between mutations in the *plc* genes to better understand the individual role of each *plc* gene in the overall phospholipase activity of *Mtb*. They concluded that *plcD* gene was more prone to insertion/deletion mutation than *plcA-C* [78]. Inactivated triple mutant *plc* exhibited a lower *plc* activity. Partial functional complementation of mutant with individual *plc* genes drew a conclusion that all 3 genes (*plcA-C*) donated equal parts to overall *plc* activity. In a recent *in-vivo* study, they postulated that though Plc's are not needed for the initiation of infection in macrophages, it is upregulated under stress like phosphate starvation conditions to maintain a persistent infection [72].

Disruption of *plcABC* or *plcABCD* impaired the growth of *Mtb* in lungs and spleen of infected mice [70]. Studies signify that *plc*-expressing *Mtb* influences necrosis rate in alveolar macrophage, i.e., cell death by suppressing PGE2 pathway. Interestingly, PGE2 content has been noted in cerebrospinal fluid of severely infected TB patients. The *plc*-deficient *Mtb* did not induce necrosis effectively, besides its ability to do so. Evading macrophage cell-death pathways by *Mtb* is seen as a strategy to avoid innate and adaptive immune responses [71]. There are still contradictions regarding the importance of mutations in inactivating *plc* activity. On treating macrophages with a PLC inhibitor (D609), reduced the levels of IP3, DAG and other inflammatory cytokines including IL-10, TNF- α during initial infection period. They also identified that lysophosphatidylcholine (LPC) (an endogenous phospholipid) directly regulated intracellular *Mtb* growth. LPC treated macrophages showed improved inflammatory cytokines secretion and better phagosome maturation [79]. siRNA-mediated knockdown studies on *plc* gene expression concluded that *plc* is involved

in the intracellular survival in macrophages and in non-replicating *Mtb* by regulating the inflammatory cytokines production [80].

Under hypoxic conditions inside an activated granuloma, where lactate and pyruvate synthesis are most likely to happen, β -oxidation and glyoxylate shunt involving isocitrate lyase (*icl*) gene pathways are actively involved in fatty acid degradation. Interestingly, the expression of the *icl* gene went up in metabolically restricted bacilli *in-vivo*. In this context, *plc* gene can be correlated with *icl* expression pattern. i.e., expression levels go up during initial infection in macrophages [81]. A study on ATP level downregulation in oxygen- and nutrient-starved *Mtb* showed that ATP level reduction is required for intact survival of non-replicating *Mtb* in the host. Their study concluded that *icl* mutants could not reduce intracellular ATP levels and eventually lost viability [81–83]. Phospholipase (*plc*) activity aids release of fatty acids from host phospholipids for glyoxylate shunt, enabling the cell to grow in fatty acid rich substrate.

Iso-citrate lyase (Icl) is a crucial enzyme expressed in the glyoxylate shunt of the tricarboxylic acid (TCA) cycle. The *icl* expression is found in *Mtb* infecting macrophages proving that, in latency and contained granulomas, the important source of carbon is likely to be fatty acids [83]. Malate synthase (MS) produced by *glcB* gene is found in granulomas and support *Mtb* latency [84]. Enzyme Icl splits isocitrate to succinate and glyoxylate and MS condenses glyoxylate with acetyl-CoA to yield malate. The Icl competes with isocitrate dehydrogenase of the TCA cycle for its common substrate isocitrate. Every cycle yields 2CO₂ molecules, enabling the cell to utilize them [83]. The *Mtb* expresses 2 genes with Icl activity, encoded by *icl1* and *aceA* (also known *icl2*). In *Mtb* strain H37Rv, *aceA* gene is cleaved into 2 isoforms namely *aceAa* and *aceAb*. However, there is disputed information about *ace* producing stable proteins or if the secreted protein have *icl* like properties [85]. Protein Icl1 is the most exploited among the two, because of its recombinant stability *in-vitro*. Isocitrate can serve as a branching point between glyoxylate shunt and citrate cycle by providing substrate for both isocitrate lyase and isocitrate dehydrogenase as shown in Fig. 1.

In a promoter trap model, *aceA* clones integrated into *Mtb* genome infected human monocytic THP-1 cell line, validated their study with the fact that the genes upregulated during infection are most likely the genes responsible for the growth and sustainability of the organism [86]. To support the earlier findings, experiments deleting both *icl1* and *icl2*, showed growth pattern similar to *icl1* and *aceA* double deletion. Further analysis by single deletion of either, had little effect on persistence [87]. This suggests that both *icl1* and *aceA* are needed for dormancy; while there can be functional complementation among the two genes. Justified observations relating antibiotic resistance to *icl*, showed evidence

that there is upregulation of Icl levels when *Mtb* is exposed to sub-lethal doses of first-line drugs [88]. Despite advancements in technology, the relation between *icl* and *glcB* is poorly understood. Regardless of intracellular *icl* levels, the level of *glcB* was not significantly altered. This behavior was seen specifically in *Mtb*. There are many proposed theories regarding this *icl* independent behavior of *glcB*, one of which is the probable shift of glyoxylate to glycine shunt pathway, where the formed succinate and glyoxylate might have roles like NAD formation for the cell in hypoxic condition [85, 89]. For long time survival of *Mtb* in host, genes *dlaT* (dihydrolipoamide acyl transferase) sub-unit of pyruvate dehydrogenase complex, *glpX* encoding fructose-1,6 bisphosphate are needed. It catalyzes the hydrolysis of fructose-1, 6-bisphosphate to form fructose 6-phosphate and orthophosphate. The product, Fructose 6-phosphate is involved in biosynthetic pathways that are important for structural components of the cell wall. The expression of later was seen in lungs of chronically infected mice and the down regulation of the same caused hindrance in gluconeogenesis and affected survival in fatty acid rich substrate [90]. This helped conclude that *glpX* could be targeted to dismantle *Mtb*'s energy cascade, since it shifts its energy source from carbon to fatty acids during chronic infection.

Phagosome acidification and blocking phagosome-lysosome fusion are two unanswered nicks in *Mtb* dormancy cascade [91]. In most enteric bacteria, glutamate dehydrogenase does ammonia assimilation directly. But since *Mtb* lacks such enzyme, extracellular glutamine and glutamate are the critical enzymes for nitrogen-metabolism. Source for glutamine are L-glutamate, ammonia and ATP from glutamate synthetase. Under nitrogen limiting conditions in host, glutamine synthesis and ammonia assimilation happens with the help of enzymes glutamine synthetase/glutamate synthetase (GS) [92]. There are 4 glutamine synthetase encoding genes (*glnA1-4*) that are post transcriptionally and translationally regulated. The assimilated glutamate is partially used by *icl* for glyoxylate shunt as shown in Fig. 1. Glutamine synthetase has control over the ammonia levels in *Mtb* infected cells and hence contributes to phagosome-lysosome fusion inhibition and phagosome acidification [69, 93]. With an interest to study the unique extracellular glutamine/glutamate synthetase, experiments involved glutamine synthetase inhibitors like, L-methionine-S-sulfoximine or antisense oligonucleotide inhibitors, showed decreased cell wall Poly-L-Glutamine/glutamate (Poly-L-Glx) and significant growth inhibition in macrophages cell culture [94, 95].

The activity of *glnA1* and *glnA2* (D-Glutamine synthetase) have been shown important for survival of *M. smegmatis* and some *Mtb* strains, while the function of other two genes are yet to be completely unveiled. Deletion mutants of each *gln* gene concluded that only *glnA1* to be essential; the mutant was an auxotroph, unable to proliferate

in macrophages. Subsequently deleting other *gln* genes, did not affect its viability [96]. At times *glnA2* is thought to synthesize D-glutamine and iso-glutamine since *glnA2* mutant was successful in infecting and causing death in mice. Therefore, *glnA2* mutant behavior was co-related with other *gln* genes compensating for its loss. Their transcriptional analysis demonstrated that L-Methionine S,R-Sulfoximine can block the activity of all four *gln* genes [96, 97]. Similar results were checked by the same research group previously, stating that the blocker excellently cut the viability of pathogenic *Mtb*, while working bacteriostatic in human THP-1 cells. They concluded that extracellular proteins can serve as targets for new antibiotics [98].

A co-infection assay examined expression profile of genes (*glnA*, *plcA* and *plcB* included) that were important for survival in macrophage, while infecting Human THP-1 macrophages and cloned them in *M.smegmatis*, it expressed more of *Mtb glnA* gene. This justified the increase in survival rates in THP-1 macrophages [99]. When THP-1 cells were cultured in 2 mM L-glutamine containing media, the intracellular L-glutamine concentration raised to 4.5 mM, which is ~ 10 times the estimated concentration. The *Mtb* growth was very good in high concentration in solid cultures while lower concentrations (1 mM-2 mM) were sufficient in liquid cultures [98]. Scaffold based inhibitors designed and validated against adenylated and de-adenylated GS, proved that they could be excellent targets based on their inhibition potential in mouse bone marrow derived macrophages [100].

The GS allows the conversion of glutamine into glutamate in the presence of ATP. For this, the intracellular GS levels are strictly controlled by *glnE* gene product, an adenylyl transferase. In stress conditions, cell responds by preserving the intracellular GS levels. Its peculiarity is that, its product inactivates GS by adenylating a conserved tyrosine residues at position -35. Hence *glnE* behaves a switch between adenylated and deadenylated forms of GS and regulates ammonia assimilation of *Mtb*. Furthermore, *glnE* promoter is highly active in ammonia/glutamine/glucose containing media [101]. The *glnA1* and *glnA3* codes for type I enzymes, regulated by *glnE*. The *glnA1*, 3 and 4 are responsible for L-Glutamine while *glnA4* codes for D-glutamine and D-isoglutamine [96, 97]. Knockout studies of vital genes responsible for cell survival, showed that *glnA1* and *glnE* mutants did not completely seize growth (required to kill *Mtb*) despite the presence of tetracycline; hence serve as a poor target [102]. The gene *glnD* (role completely not known) also has a part in deactivating the GS by uridylation. Deletion of *glnD* would prevent the uridylation of its substrate, PII protein (secreted by *glnB*). Unmodified PII protein promotes adenylation of *glnE*, thus inactivating *glnA*. Read et al. noted despite lowering GS levels in macrophages, *glnD* mutant SCID mice overall survival of was not affected [103]. They concluded that *glnA* prevalence even at lower rates are

sufficient for the survival of the organism, however the role of *glnA3* that is also under the control of *glnE* could not be cut off completely.

Oxidative stress response

Availability of iron is a critical infection factor for *Mtb*. It has a sophisticated iron-acquisition system to combat iron depletion imposed by host. Its core lies in production of extracellular secreted small molecules (siderophores) to bind tightly and internalize available iron. Iron is a co-factor for redox reaction enzymes and other vital functions without which the bacterium shows stunted growth. Early reports prove that human serum is tuberculostatic in nature unless a reversal is done by addition of iron [104]. However, human serum has iron content that is in tight relation with iron binding proteins and hard for bacterial uptake as such [105, 106]. The *Mtb* requires iron in non-heme form for biosynthesis of amino acids, pyrimidines, TCA cycle and electron transport chain enzymes [107].

In response to the stress created, these iron chelators (siderophores) named mycobactin takes iron from the host. The siderophore has two different forms, one of which is hydrophilic and the other hydrophobic. The hydrophilic mycobactin (Carboxymycobactin) is secreted externally, binds to available iron and competes with the host's iron binding molecules. The other hydrophobic group of mycobactin carries the bound iron through the extremely hydrophobic cell envelope. The difference between the two classes of mycobactin is believed to be the modification in salicylic acid moiety [108]. The mycobactin internalization is assisted by a small trans-membrane (histone-like) protein HupB coded by *hupB* gene [109]. Mycobactin is needed for multiplication inside the macrophage microenvironment. Although iron is an important requisite, equally important is the tight regulation of iron, which would otherwise be lethal due to accumulation and production of oxygen radicals. The *Mtb* has 4 genes whose transcription levels are modified to negatively regulate iron uptake [108, 109]. The genes *furA-B* encodes proteins of Ferric uptake regulator class (Fur), while *DtxR* (diphtheria toxin repressor) family accommodates genes *ideR* and *sirR*. The *DtxR* (siderophore class molecule in *Corynebacterium diphtheria*) regulates diphtheria toxin gene. Due to its well derived structure and function, it is considered prototype for such regulatory proteins. In *Mtb*, *IdeR* shows a striking 88-90% similarity in the first 180 amino acid domain with *DtxR*. They share a highly specific helix-turn-helix domain crucial for palindromic DNA binding. The prominent iron-dependent siderophore role of *IdeR* was first identified in *M.smegmatis* [110]. A cluster of 10 genes (designated *mbtA-J*) that produces enzymes for assembly of mycobactin was identified in a study. Five of the genes (*mbtB-F*) formed the peptide synthetases and

polyketide synthetases to elongate mycobactin core. The *mbtA* provides activated salicylate while *mbtB* utilizes the same to do chain initiation. When replacing *mbtB* gene with an antibiotic marker, it resulted in a mutant *Mtb* strain with no mycobactin and restricted growth under low-iron environment. Additionally, their study has given evidence for intersection of *fad* genes with *mbt* genes and *fad* genes are iron repressible [107, 111]. Mutation studies on *mbtB* showed stunted growth in the human macrophage cell line THP-1. Early experiments were done to find if *ideR* affects the transcription of *dtxR*-regulated promoters. DNA mobility shift assays and DNA foot printing analysis revealed that IdeR and DtxR proteins binds in an iron dependent manner, to the same promoter sequence [112]. Crystalline structure analysis of IdeR and DtxR revealed that they are homodimeric proteins in their active state forming a dimer-dimer DNA binding complex finding itself to the operator region. An *ideR* monomer has 3 domains and 2 metal ion binding sites [113]. In abundance of iron, *ideR* binds upstream to promoter responsible for iron uptake (*mbt* genes) and represses it. This activates *bfrA* (bacterioferritin), *bfrB* (ferritin) (Iron storage proteins) and *katG* (catalase peroxidase) genes [114].

Under Iron deficiency, *ideR* is not available to bind to the promoter thus expressing the proteins responsible for oxidative stress repression and iron acquisition. On deletion of specific cassettes in *Mtb ideR* gene, impaired the growth in human THP-1 cells concluding the strong relationship between siderophores and virulence in *Mtb* [115]. Conditional mutant of *ideR* proved its requisite for *Mtb* growth *in-vitro* and *in-vivo*. Hit compounds inhibiting *ideR* were generated since it has huge control on *Mtb* metabolism. Pharmacophore based screening, yielded about 9 lead compounds that work well against *ideR* target [116]. Chou et al., (2004) demonstrated in a fluorescence assisted tracking of Fe^{2+} binding to *ideR*, showed the importance of dimerization of the Fe^{2+} towards binding site [113]. In a synthetic drug designing study, 2 compounds were found to increase MIC, implying that the drug are active against *Mtb*. Results were confirmed by RT-PCR [117]. Fully activated structure of IdeR was described by Pohl et al., (1999) at first and then additions in the structure were described by other groups [118]. The latest findings suggest the possible interaction between *ideR* and *mbt* gene regions. A similar study on DtxR/tox binding, complete structure was reported by Schmitt et al., (1996) [112]. Manabe et al., (2005) in their study showed evidences about mutant hyper repressor impairing the ability of *Mtb* to upregulate mycobactin synthesis [119]. Additionally, some studies show the presence of active Zn^{2+} molecule in its pocket. Postulations that urease operon and ammonia generated by the urease activity and other transcription regulatory protein factors like *Rv1404* and *Rv1847* are needed for iron assimilation [120]. Early findings about complete sequencing of *ideR*

gene with *mbtB* and *mbtE-F*, which are put forward as peptide synthetases, show that it can form an excellent target for selective inhibitors to treat tubercular infections. Despite the abundance of iron in the medium, *ideR* mutants had stunted growth. However expression and working of *bfr* genes were not affected [115].

An attractive pathway to control metabolic activity and cease the growth of *Mtb* is by inducing stress through a variety of factors like aging logarithmically growing cultures by oxygen limitation. The *acr* (α -crystalline) (Also known as *hspX*) gene encodes a 16 kDa heat shock/chaperon proteins [121]. Initial perceptions state Acr to be a key membrane protein but later they went on to reveal that Acr is a chaperon with oligomeric active sites consisting trimer of trimers [122]. Over expression of Acr protein is associated with delayed aging and slower growth rate. Therefore, it is invoked as an important factor in latent TB infection. Studies reveal the upregulation of *acr* gene is found in cultures with limited oxygen supply or nitric oxide (NO) rich microenvironments [123, 124]. The Acr belongs to a family of dormancy regulon proteins, DevR (also called DosR). The *dosR* regulon controls other genes involved in fatty acid metabolism, energy acquisition and nitrate reductase which include *narX*, *nark2* (nitrate accumulation and electron transport) and *nrdZ* (dNTP synthesis under hypoxia) that are activated in response to hypoxia or depleting nitric oxide [65]. The MprA-B is a two component system similar to DevR associated with stress response proteins regulation which is triggered during low nutrient content and SDS. The MprA-B disruption showed a sudden hike in other stress response proteins including DosR and SigD. The MprA binds upstream to *sigB* and *sigE* *in-vitro* and regulates the expression of *sigB* and *sigE* [125]. Additionally *ideR* regulons including *rpf*, *mmpl*, *mbtB*, *hisE* (iron response genes) were activated. Gene *mprA* regulates its own self while interacting with *acr2* promoter and controls the house keeping gene *sigA* [126].

Base substitution analysis of *acr* genes and its promoter roles concluded that regulation of *acr* and its promoter is heterogeneously regulated at multiple levels [127]. Another study on gene regulation of 8 gene promoters that included *acr*-coregulated promoter (ACP) family, demonstrated both in computational and experimental approaches concluded that standing cultures with low oxygen content supported the best for *Mtb* growth. Up-regulation of heat shock proteins were seen 24 h post infection in THP-1 cells [128, 129]. An *in-vitro* study concluded that defined oxygen concentrations where *acr* transcription was strongly induced by mildly hypoxic conditions and macrophage infection, improved *Mtb* survival [130]. In a whole genome microarray, > 100 genes that are instantly altered with hypoxia were identified, many of which are important in cell component biosynthesis and aerobic metabolism [131]. In a recent study attempting to

relate the structure and function of oligomers of *acr* genes, chromatographically purified recombinant proteins were studied for chaperon activity based on its binding to insulin [132]. Despite considerable research, mechanism of *acr* and role in evading immune response in *Mtb* is unknown. The *acr* gene is an emerging drug target since it interacts with other groups like *ideR*, *mmpL*, *sig* genes of *Mtb* that help maintain homeostasis and survival.

New drug targets for latent TB infection

From the early identification and practice, some drugs work well against actively replicating bacteria but are not useful in sterilizing dormant *Mtb*. Dormant *Mtb* has no correlation between MIC (minimal inhibitory concentration). INH for example, is active against growing bacteria but does not work on latent organisms. Pyrazinoic acid (POA) works poorly on growing bacteria but is excellent in sterilizing non-replicating organisms [133]. For the existing complication in TB treatment, warranting new drug identification and validation of appropriate targets are urgent. Since the physiological state of latent *Mtb* is unknown, owing to the hypothesis that the organism shifts from growth and replication related metabolisms to persistence and survival, a drug target intrinsic for both active and dormant *Mtb* is desirable allowing the treatment for both clinical conditions. Conceptions state that drug targets required for prolonged *Mtb* survival may be similar to those needed for its persistence during prolonged TB chemotherapy. Equally important opportunities must be given to drug repurposing to establish activity on wide target range. Also, an alternative currently emerging approach for latent TB treatment is the reawakening of persistent organism by inhibiting host immune response or targeting dormancy pathways enabling the organism susceptible to conventional drugs. Among the current discussed genes some are already well known targets, some are drug candidates and some are proposed targets for potential drugs.

The cell wall synthesis enzymes are absent in humans, representing primary targets for anti-TB drugs. Among discrete fatty acid synthesis enzymes, FadD32, MabA and Inh serve as drug targets. Protein MmpL-3 is a drug target for SQ-109, an ethambutol derivative which is in phase-II trials currently. It works well on ethambutol resistant *Mtb*, confirming the difference in its mechanism [134, 135]. Genes *cma* and *mma* are valuable targets for FAS-II inhibitors, thi-acetazone analogs like SRI-224 and pro drugs NAS-21 and NAS-91 [136].

Glyoxylate shunt enzymes such as malate synthetase (MS) and Icl are good targets since they are required for persistence of *Mtb* in macrophage. However the crystal

structure of MS is repositied and new drug agents are yet to be discerned.

Regulatory proteins like GlnD, GlnE, AhpE (family of Ahp proteins) and IdeR are required for the persistence of *Mtb* in macrophages and hypoxia, serving them as emerging drug targets. Bedaquilin is a successfully emerging Anti-TB drug blocking ATP synthase. Q203 is in phase-I trials blocks respiratory cytochrome-b subunit and interferes with the iron—sulfur transport protein, depleting intracellular ATP. It works both in aerobic and anaerobic conditions on replicating and non-replicating bacteria [137].

Widely, homology modeling and docking studies were employed to find the difference in binding sites of KasA and KasB that are targets for Thiolactomycin (TLM). Promising results were shown when there was conformational change upon TLM binding in KasB. Another drug platensimycin targets Kas proteins of the FAS-II system in active *Mtb*, while its analog platencin is active against both active and dormant *Mtb* [138, 139]. Owing to its importance in persistence, other genes *pcaA* and *devR* can also be proposed as drug targets. Table 1 comprises the proposed gene targets for new and repurposed drugs and their inter-relation in the infection cascade of latent *Mtb*.

Conclusion and perspectives

Though there are many well-known strategies of persistence and mechanism of *M. tuberculosis*, it continues to post challenge on deriving successful methods for complete elimination from the host. The mechanism with which it modulates the host immune system is to be resolved. While there are advancements in the field regarding the diagnosis and prediction of its mechanisms facilitating the forward push to decode its functional base, like Sassetti, [142] demonstrated an *in-vivo* study with mutations in all the genes of *Mtb*, identifying 194 genes involved in pathogenesis. A major section of the screened genes, were involved in lipid metabolism and transportation of inorganic compounds and carbohydrates across the cell wall. Though there are innumerable genes for pathogenesis, survival of *Mtb* within the macrophage activates only a certain set of genes relating cell wall modulation and lipid metabolism. Analyzing these genetic factors and deducing their interaction would give away details on its resistance pattern and idea about novel targets that can be targeted to control its pandemic. The virulence factors described in this review focuses on the genes modulated by *Mtb* to enable its survival inside an activated macrophage. The detailed explanation on genes and proteins accounting for cell entry, Fatty acid degradation, mycolic acid synthesis, phagosome acidification inhibition, inhibition of phagosome-lysosome complex, iron uptake, oxidative stress resistance and chaperon protein modulation are

Table 1 List of genes responsible for *Mtb* survival and their drug candidate status

S. No	Gene	Abbreviation	Function	Interrelation	Drug candidates	References
1	<i>mce</i>	Mammalian cell entry	Invasion, fatty acid degradation, ABC transporter	<i>mce</i> works with <i>fad</i> gene for lipid import (ABC transporter) and degradation		[19, 22]
2	<i>icl, ace</i>	Iso citrate lyase	Survival in granulomas and latency, glyoxylate shunt pathway enzyme	Takes up glutamate from <i>gln</i> gene product (ammonia assimilation)	Proposed new target	[81, 85]
3	<i>glcB</i>	Malate synthase	Glyoxylate shunt pathway enzyme		Proposed new target	[84]
4	<i>cma</i>	Cyclopropane α -mycolic acid	Mycolic acid subunits synthesis	Works with <i>fad</i> and <i>inhA</i> gene for cell wall generation		[3]
5	<i>mma</i>	Methoxy mycolic acid	Involved in α -mycolate synthesis	Cell wall mycolate component (fatty acid degradation process)	Pro-drugs NAS21 and NAS91 proposed	[61, 140]
6	<i>pcaA</i>	Cyclopropane mycolic acid synthase-III	Cyclopropane ring formation, Cell envelope permeability, persistence	Involved with <i>mma</i> gene in cyclopropanation of the proximal rings	Novel identified target	[61, 134]
7	<i>ideR</i>	Iron dependant repressor	Iron uptake/ regulation, siderophore synthesis	Involved with <i>katG</i> , <i>ahpC</i> in detoxifying from free radicals	Emerging drug target	[19, 141]
8	<i>fur</i>	Ferric uptake regulator	<i>ideR</i> gene regulation	Involved with <i>ideR</i> in its regulation		[108]
9	<i>katG</i>	Catalase peroxidase	long chain fatty acid inhibition, inactivates <i>inhA</i>	Inactivates <i>inhA</i> to stop mycolic acid production	Known target for INH, proposed re-emerging target	[48]
10	<i>inhA</i>	Enoyl acyl carrier protein	Mycolic acid synthesis	Works with <i>fad</i> in mycolic acid subunits synthesis	Existing target for INH, proposed repurposed target	[133]
11	<i>fad</i>	Fatty acid degradation	Cholesterol metabolism	Acetyl-CoA donated to fuel glyoxylate shunt, Mycolate synthesis with <i>cma</i> , <i>inhA</i> and <i>mma</i>	Proposed target	[27, 134]
12	<i>kstR, sigE</i>		Fatty acid degradation regulation	Works with <i>fadE</i> gene and regulates it		[32]
13	<i>gln</i>	Glutamine synthase I	Ammonia assimilation, glutamine synthesis	Glutamine is supplemented to glyoxylate shunt and fatty acid metabolism	Q-203 – phase I trials, Known target for bedaquillin	[92, 141]
14	<i>acr</i>	α -crystallin	Inhibition of phagosome lysosome complex, phagosome acidification	Chaperon protein and maintains cell wall homeostasis	Proposed drug target	[121, 127]
15	<i>plc</i>	Phospholipase	Fatty acid degradation	Used in β -oxidation cycle with <i>fad</i> and in close relation with <i>icl</i> gene		[70, 71]
16	<i>ahpC</i>	Alkyl hydro peroxide reductase	Substitutes <i>katG</i> function in some cases	Involved with <i>katG</i> gene and plays detoxifying role		[53]
17	<i>oxyR</i>		Tightly regulates <i>ahpC</i> gene			[53]
18	<i>mmpL</i>	Mycobacterial membrane protein large	Translocates cell lipids across the membrane	Related to <i>katG</i> in combating cell stress	SQ-109 Phase –II trials	[134]
19	<i>kasA/B</i>	Keto acyl CP synthase	Precursors of mycolic acid	Works with efflux pump genes to create resistance	Known target for INH, repurposed target for thiolactomycin and platencin	[134]
20	<i>DevR</i>	DNA-binding transcriptional activator	Regulates fatty acid synthesis and energy metabolism genes	Related to <i>fad</i> and <i>hspX</i>	Proposed new target	[134]

described in this review. The summarization on the roles and interaction between these genes is expected to provide clear insight on vaccines development, drug targeted therapy and resensitizing *Mtb* cell to pre-existing drugs.

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